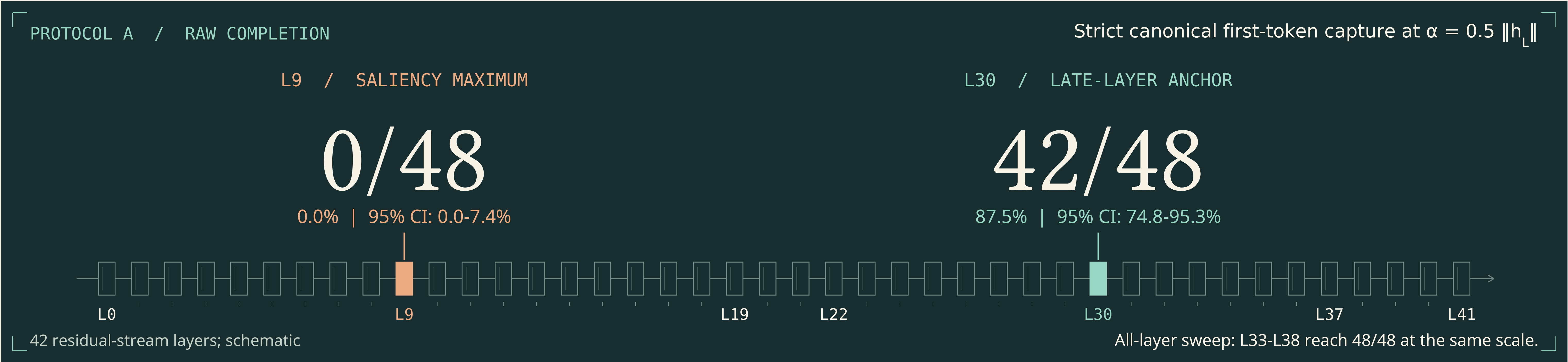


From Saliency to Actionability: Layer Localization Is Protocol-Conditional

Saliency nominates a layer. Intervention must validate the choice.

Behnood Mashayekhi · Arian Amiramjadi · Parsa Hosseini

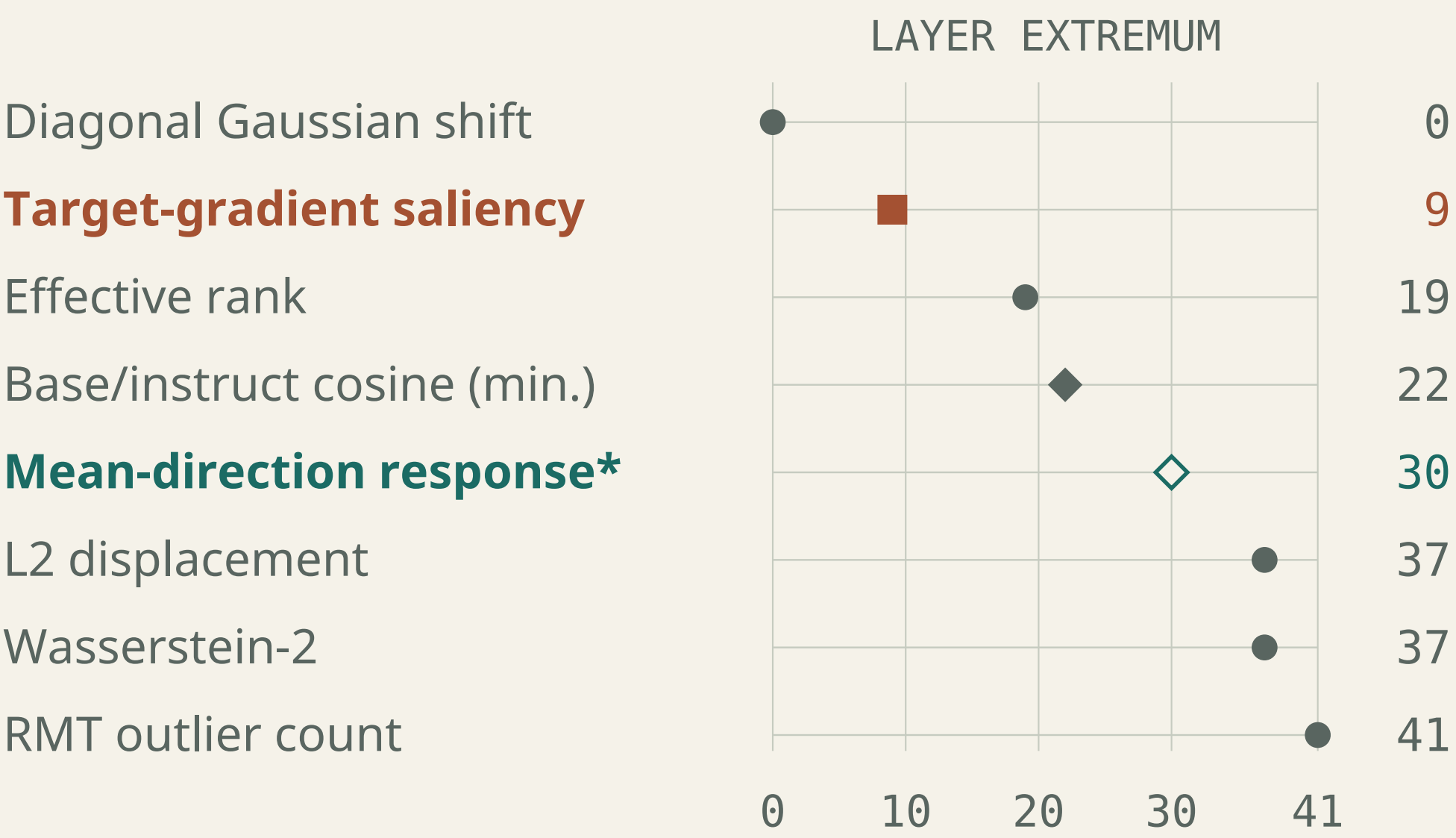
Gemma-4-E4B / Gemma-4-E4B-it | 42 layers | 48 probes



01 / LOCATE

The measurement changes the location.

Eight measurement objects in the same base-to-instruct study (Table 2).

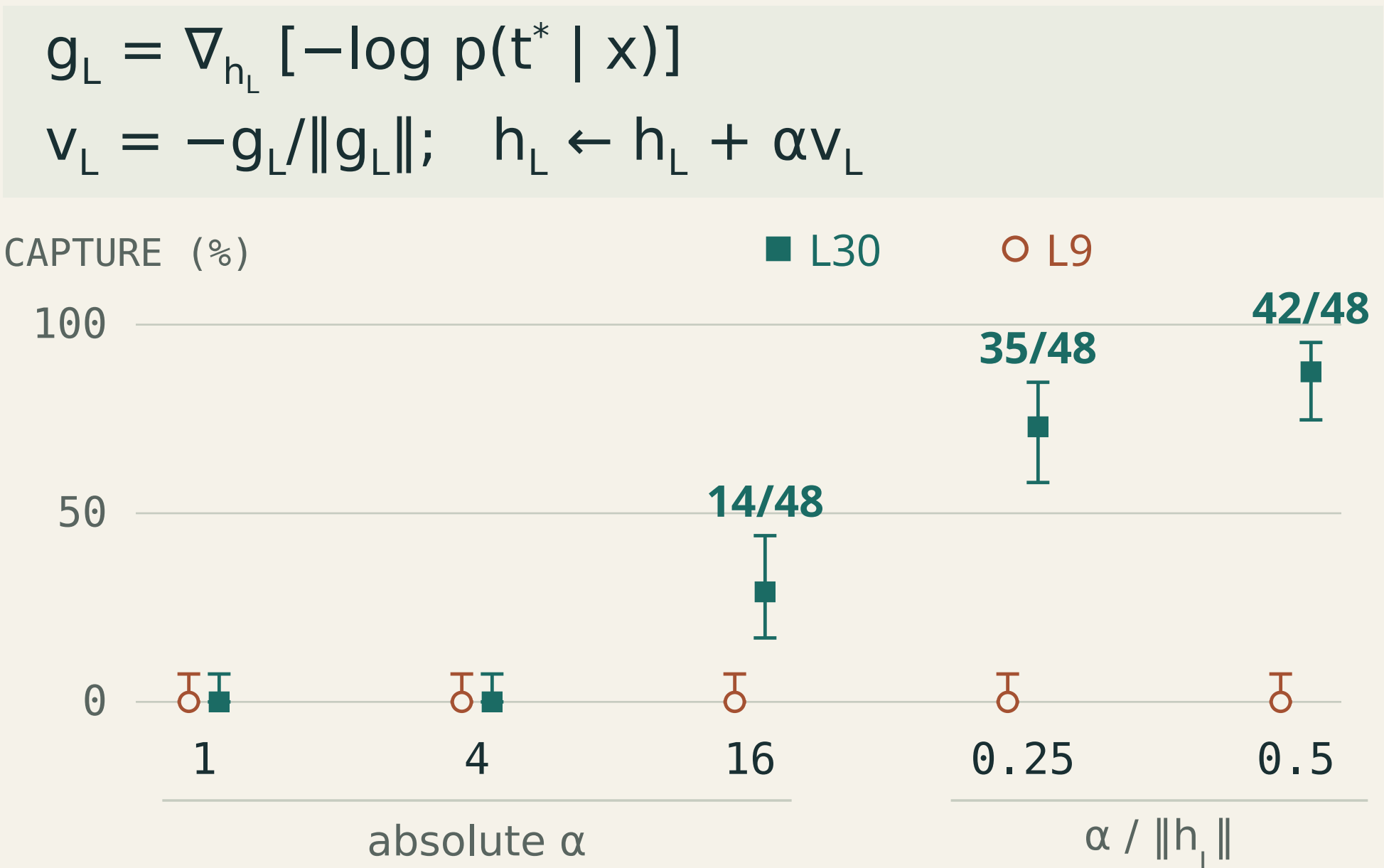


*Largest among five sampled layers. Diagonal shift is variance-normalized; it is not a full Fisher estimate.

02 / INTERVENE

The patch has to pass its own test.

Instruct model only; last prompt token. t^* is the oracle canonical first answer token. Saliency ranks raw gradient magnitude.

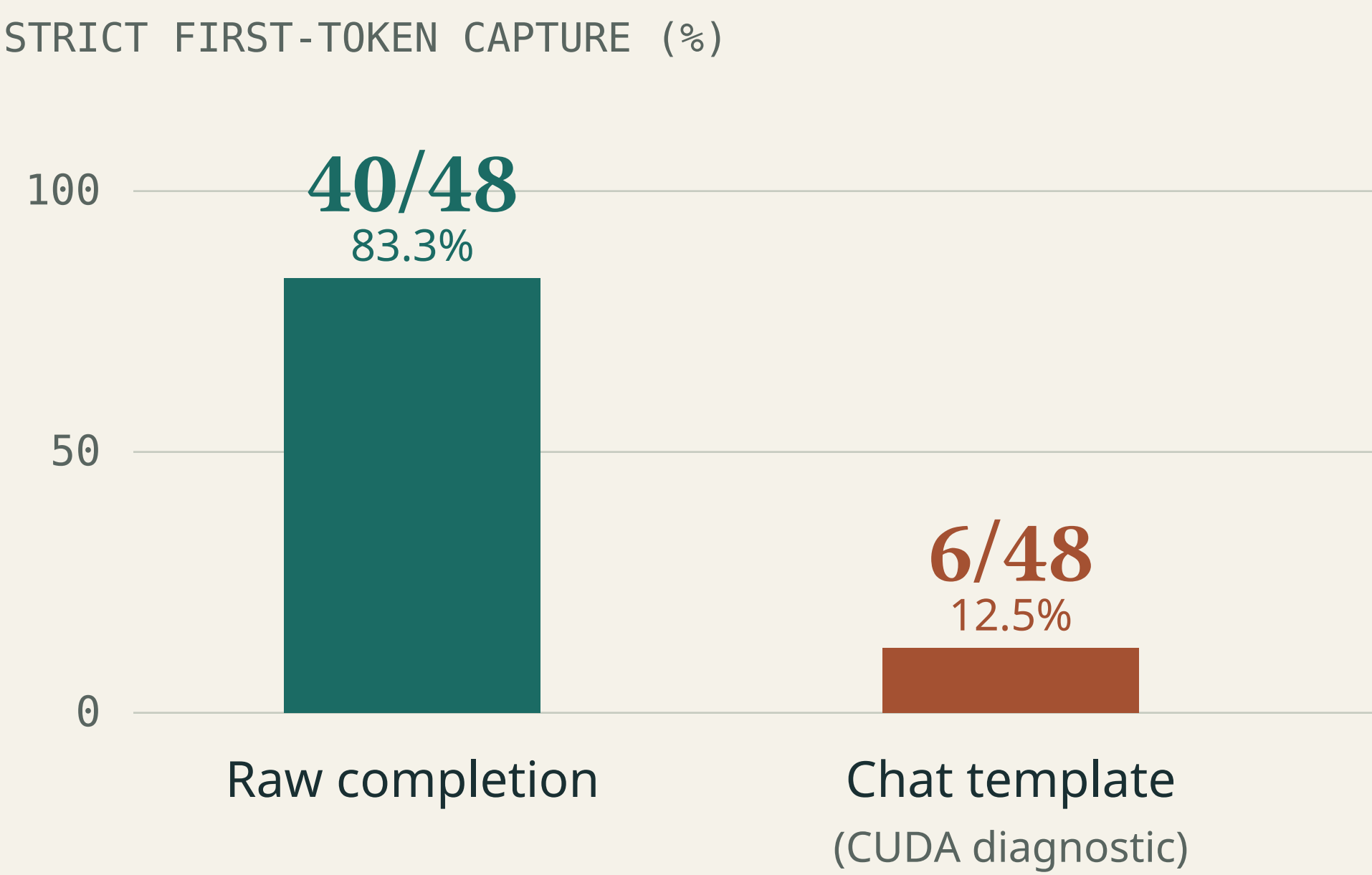


L9: 0/48 at every tested scale. Whiskers: pointwise exact 95% intervals; no correction for layer selection.

03 / CHANGE THE PROMPT

Same layer. Different prompt mode.

Matched Protocol C at L30. $\alpha = 0.5 \parallel h_L \parallel$; 48 probes per condition.



Separate MPS chat diagnostic: 1/48. Device/protocol results are not pooled.

04 / REPORT THE COMPLETE CLAIM

A layer number needs its experimental conditions.

CONTEXT	MEASUREMENT	INTERVENTION	VALIDATION
Model + contrast	Object + layer	Direction + patch	Outcome criterion
Prompt mode	Selection rule	Scale	Uncertainty
Task context			

First-token capture tests prefix control, not answer quality.

What remains unestablished

Directional specificity and general-purpose steering.

One model pair; 48 aggregate probes. Matched sign-flip, random-gradient-direction, wrong-target, task-stratified and held-out controls remain open. Exact checkpoint revisions are not retained.

L9 is not inert: target log-probability rises by 3.19 on a separate 40-probe subset.